

Vision Language Model Comparison

Performance benchmarks across model sizes, formats, and precision levels

Model Name	Format / Precision	Est. RAM Usage	Vision Processing Time (TTFT)*	Text Gen. Speed (Tokens/Sec)
qwen3.5-vl-0.8b	GGUF - 4-bit (Q4_K_M)	~700 MB – 900 MB	1 – 2 Seconds	40 – 60 t/s
qwen3.5-vl-0.8b	GGUF - 8-bit (Q8_0)	~1.1 GB – 1.3 GB	1.5 – 2.5 Seconds	32 – 48 t/s
qwen3.5-vl-0.8b	Non-GGUF - 16-bit (FP16)	~2.0 GB – 2.5 GB	3 – 5 Seconds	12 – 20 t/s
qwen2-vl-2b	GGUF - 4-bit (Q4_K_M)	~1.6 GB – 1.9 GB	2.5 – 3.75 Seconds	20 – 32 t/s
qwen2-vl-2b	GGUF - 8-bit (Q8_0)	~2.5 GB – 2.8 GB	3.75 – 5 Seconds	12 – 20 t/s
qwen2-vl-2b	Non-GGUF - 16-bit (FP16)	~4.5 GB – 5.5 GB	6 – 8 Seconds	4 – 8 t/s
qwen3.5-vl-2b	GGUF - 4-bit (Q4_K_M)	~1.6 GB – 1.9 GB	2.5 – 3.75 Seconds	20 – 32 t/s
qwen3.5-vl-2b	GGUF - 8-bit (Q8_0)	~2.5 GB – 2.8 GB	3.75 – 5 Seconds	12 – 20 t/s
qwen3.5-vl-2b	Non-GGUF - 16-bit (FP16)	~4.5 GB – 5.5 GB	6 – 8 Seconds	4 – 8 t/s
moondream2-2b	GGUF - 4-bit (Q4_K_M)	~700 MB – 900 MB	1.5 – 2.5 Seconds	35 – 50 t/s
moondream2-2b	GGUF - 8-bit (Q8_0)	~1.1 GB – 1.4 GB	2 – 3 Seconds	25 – 35 t/s
moondream2-2b	Non-GGUF - 16-bit (FP16)	~3.2 GB – 3.8 GB	4.5 – 6 Seconds	8 – 14 t/s

* TTFT = Time To First Token. Values are estimates and may vary depending on hardware, OS, and model loader configuration. moondream2-2b figures are estimates scaled proportionally from its known lightweight architecture (compact vision encoder, weight-shared decoder) and publicly reported ~1.2GB 4-bit RAM footprint; not directly measured on this hardware.